Data resources: from records to a reproducible benchmark

Paper section 3.2: *Data repositories for system identification and regression*

Seven resource families named in the article

DaISy
Dynamic identification records

IFAC
Identification benchmark records

NIST
Statistical reference data

MATLAB
Toolbox and demonstration data

UCI
Machine-learning data repository

DELVE
Learning-method evaluation data

WEKA
Benchmark data collections

These are sources of experiments and regression tasks, not seven physical process models. Availability, licensing and original links must be verified for each selected record.

One selected source record

1. Acquire + identify
Source URL, license, version, checksum and apparatus description

→

2. Import + inspect
Time base, input/output order, units, missing values and operating range

→

3. Freeze the split
Separate fitting, selection and final evaluation records before estimation

→

4. Compare models
Same samples, excitation, horizon, metrics and uncertainty conventions

Dynamic identification

Keep temporal order and independent runs. Prevent overlap of lagged regressors across split boundaries; state the sample period and prediction horizon.

Static regression

State the target, feature units and sampling unit. Preserve grouped dependence. Use a declared validation split and a final held-out test.

Shared safeguards

Fit normalization only on training data. Record exclusions, provenance and random seeds. Report baseline errors with the same evaluation contract.

This resource map explains Section 3.2 and the repository import workflow. It does not claim that every historical third-party data set is bundled or still downloadable. Dataset-specific provenance remains necessary before numerical comparison.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.